Leveraging Genre-Based Music Preferences to Assist Psychologists in Tailoring Mental Health Treatments : A Machine Learning Approach

A research project submitted in partial fulfillment of the requirements for the Bachelor's degree in Computer Science and Engineering

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APPROVAL

The research project report titled "**Leveraging Genre-Based Music Preferences to Assist Psychologists in Tailoring Mental Health Treatments : A Machine Learning Approach**" submitted by **Mohammad Mahbubur Rahman** (ID: CE-20001) and **Syed Mahir Faisal** (ID: CE-20005), Session 2019–2020, to the Department of Computer Science and Engineering, Mawlana Bhashani Science and Technology University, Tangail-1902, Bangladesh, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of **Bachelor of Science (Engineering)** in Computer Science and Engineering and approved as to its style and contents.

Board of Examiners

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DECLARATION

We, **Mohammad Mahbubur Rahman** and **Syed Mahir Faisal**, hereby declare that the task presented in this thesis is the outcome of the investigation performed by us under the supervision of **Dr. Md. Sazzad Hossain**, Professor, Department of Computer Science and Engineering, Mawlana Bhashani Science and Technology University, Santosh, Tangail-1902, Bangladesh. We further declare that no portion of this thesis or its components has been or is being considered for submission elsewhere in order to receive a degree or diploma.

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Abstract

In this paper, we focuses on leveraging genre-based music preferences, which are determined by individuals' listening frequency of certain genres, to assist psychologists in tailoring mental health treatment using various machine learning algorithms. Various studies were done previously, which incorporate music and emotional well-being, but either fail to build a personalized genre-based music recommendation system for mental health patients or depends on complex data, i.e., different physiological signals, which requires the use of costly sensors. Our study aims to incorporate between-persons' genre-based music listening behavior with their mental health well-being and decide whether the frequency of listening to a certain genre of music may or may not improve their condition. The data we used focuses on the musical effect on person's with four different mental health conditions, which are depression, anxiety, insomnia, and OCD, and how much they listen to different genres of music. During our research, we compared three different models and found out that XGBoost gave the best results with accuracy reaching upto 94% and macro F1-score reaching upto 91%. Various Exploratory Data Analysis in our study demonstrated how person's favorite genre is connected to their mental health conditions. Our approach holds promise for giving personalize music therapy to mental health patients based on their genre based music listening frequency. Additionally this study, works with individuals' listening frequency of 16 different genres of music, which ensures that it focuses on their overall music listening habit.

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List of Abbreviations

NN	Neural Network
XGBoost	Extreme Gradient Boosting
WHO	World Health Organization
RF	Random Forest
SMOTE	Synthetic Minority Oversampling Technique
EEG	Electroencephalogram
GSR	Galvanic Skin Response

Chapter 1

Introduction

1.1 Background

According to recent global estimates, over 970 million people worldwide are living with a mental health disorder as of 2025 [10]. This includes conditions such as depression, anxiety, bipolar disorder, schizophrenia, and others. Personalized music therapy for patients with these conditions can have a significant impact on mental health treatments.

Anxiety, depression, insomnia, and obsessive-compulsive disorder (OCD) are all mental health problems that are now big problems around the world that affect millions of people's health. Cognitive-behavioral therapy and medication are still the best ways to help with these issues. But therapies that use music, for example, are getting more and more popular because they can help with mental health without hurting or costing a lot of money.

People have a special connection to music. It can be fun and help them deal with their feelings and get mental health support. Studies have shown that the music a person likes can both show and change their mental health [1]. For example, classical or upbeat music is often associated with improved mood regulation, while melancholic or intense genres may offer emotional release for some individuals yet potentially worsen symptoms for others, depending on their emotional state and context. Despite growing awareness of these nuanced effects, many current music therapy practices remain generalized and overlook the importance of tailoring interventions to individual preferences grounded in empirical data.

1.2 Motivation

The growing integration of technology into mental health care is reshaping how individuals understand and manage their emotional well-being. With the rise of digital platforms, people now routinely monitor their mental health through mobile apps, self assessment tools, and online check-ins—turning personal well-being into a more trackable, day-to-day experience. At the same time, music streaming services are quietly recording patterns in listening behavior, from favorite genres to daily mood-based choices. Together, these digital traces offer a powerful opportunity to uncover how music preferences reflect and influence psychological states, paving the way for more personalized and adaptive mental health interventions.

Despite the well documented emotional impact of music, therapy treatments that use it frequently lack personalization and are not empirically supported. Traditional music therapy methods usually rely on therapist intuition or broad genre associations, without taking into consideration individual mental health profiles, listening habits, or contextual elements that influence musical receptivity. This leads to wasted opportunities to adapt interventions that are more meaningful to each patient.

Machine learning offers a powerful framework to bridge this gap. By analyzing correlations between self-reported mental health scores and frequency of listening to specific genres, predictive models can be developed to anticipate the likely psychological effects of certain musical inputs whether they lead to emotional improvement, stability, or deterioration. Such evidence-driven insights can help psychologists move from trial-and-error strategies to more confident, data-informed recommendations.
The motivation behind this research is twofold. First, it aims to empower individuals by offering them music-based interventions that align with their unique mental health needs. Second, it seeks to assist mental health professionals by providing intelligent systems that generate personalized therapeutic suggestions, thereby enhancing clinical decision-making. In a world where mental health support is increasingly mediated by digital tools, this approach contributes to the growing body of work focused on ethical, personalized, and data-driven mental health care.

1.3 Problem Statement

Music therapy holds significant potential as a complementary approach to mental health care. However, because it relies on standardized methods that ignore individual variances, its practical application often falls short of expectations. Many interventions use uniform playlists or general frameworks for everyone [7], disregarding the personal and emotional connections people form with different types of music. How a song can impact an individual depends on so much as mood, life experiences, and even the culture one grew up in. For some, a genuine classical piece might lift their spirits; for others, a heavy or rock track could either give them relief or drag them deeper into distress. Using identical music interventions across the board can, frankly, fall short of helping and, in some cases, might even make things worse emotionally. As healthcare shifts toward more personalized, evidence-driven practices, there's a clear need to leverage tools like machine learning to dig into how someone's music preferences tie into their mental well-being. Harnessing data enables us to shape therapeutic experiences that genuinely reflect an individual's emotional landscape, focus on how specific music genres or songs influence an individual's emotional state, whether they provide comfort, trigger distress, or simply pass without much effect. These insights can elevate music therapy, allowing clinicians to offer treatments that align more closely with each patient's emotional needs and psychological profile.

1.4 Research Questions

This research is anchored in the belief that music when thoughtfully aligned with an individual's psychological profile can serve as a meaningful component in mental health care. However, realizing this potential requires a data-informed understanding of how different music genres influence mental well-being across diverse individuals. With this goal in mind, the study is guided by the following core research questions:

- Can individual variations in mental health-measured through self-reported scores of conditions such as anxiety, depression, insomnia, and OCD—combined with patterns in music listening behavior, be used to reliably predict the psychological impact of listening to specific music genres?
- Among the machine learning models explored, namely Random Forest, XGBoost, and Neural Networks, which approach demonstrates the highest level of accuracy, consistency, and interpretability in predicting whether the frequency of listening to a certain music genre will have a positive, neutral, or negative psychological effect?
- How can the predictive insights generated by this framework be translated into actionable recommendations for clinicians, therapists, or mental health platforms? This final question focuses on the practical application of the findings, examining how genre level predictions can be incorporated into personalized, non-invasive treatment plans that respect individual listening preferences and psychological needs.

1.5 Research Objectives

This study aims to build a predictive machine learning framework designed to assess how listening to different music genres may affect an individual's psychological state. By leveraging self-reported mental health scores alongside the frequency of genre-based listening behavior, this research explores the potential for data-driven personalization in music-based interventions. The ultimate objective is to bridge the gap between empirical data and clinical application, offering psychologists a more nuanced, evidence-based tool to integrate music meaningfully into mental health care.

The specific objectives of this research are:

- To collect and preprocess data related to individuals' self-reported mental health indicators, including anxiety, depression, insomnia, and obsessive-compulsive disorder (OCD), and their listening frequencies across a range of music genres.
- To restructure the dataset into a machine learning-compatible long format, enabling each genre-listening instance to be modeled independently, and to encode the psychological effect of music (i.e., Improve, No Effect, or Worsen) as the predictive target variable.
- To train and evaluate three supervised classification algorithms—Random Forest, XGBoost, and Neural Networks on their ability to predict the probable psychological effect of a given music genre for individuals based on their mental health profile and listening frequency.

- To generate genre-specific predictions that can assist psychologists in identifying which listening behavior may be helpful, neutral, or potentially counterproductive for individual patients, thereby contributing to more personalized, non-pharmacological treatment strategies.

1.6 Organization of the Thesis

This thesis is organized into six chapters:

- Chapter 1 – Introduction: Establishes the context, significance, and aims of the research.
- Chapter 2 – Literature Review: Summarizes existing work in music psychology, machine learning in mental health, and highlights the current research gap.
- Chapter 3 – Methodology: Describes the dataset, data transformation strategies, preprocessing pipeline, and modeling procedures.
- Chapter 4 – Results and Discussion: Presents the outcomes of model evaluations and interprets the implications of genre-based predictions.
- Chapter 5 – Conclusion: Summarizes findings, discusses limitations, and proposes directions for future work.

Chapter 2

Literature Review

Numerous studies have demonstrated the impact of music on our mental health, which is generally considered universal. It has been shown to affect our mental health, stress levels, mood, and cognitive abilities. There are different types of music as genres. Big data and machine learning have made it feasible to link listening habits to mental health conditions and music genres, as well as to use music therapy for those suffering from mental health problems. An overview of the major works of literature that form the basis of this thesis is provided in this chapter.

2.1 Existing Methods

Several methods were introduced in previous studies that focus on the connection between individuals' emotional well-being and music. Some of these methods are:

2.1.1 Random Forest Classifier

Random Forest is a popular team-based learning approach that combines multiple decision trees. Each tree is trained on random chunks of data and features, and they work together by voting to decide the final classification outcome. Karunya et al. [3] and Andrew et al. (2024) [4] implemented random forest in their research.

2.1.2 Artificial Neural Networks (ANN)

Artificial Neural Networks are made up of layers of connected nodes, like neurons in a brain, working together to uncover complex, non-straightforward patterns in data.Rahman et al. (2020) [2] trained a neural network using biometric features.

2.1.3 K-Nearest Neighbors (KNN)

K-Nearest Neighbors, or KNN, is a straightforward method that sorts data points by looking at their closest neighbors in a dataset. It doesn't rely on strict assumptions about the data's structure, making it flexible. Essentially, KNN decides how to classify something by checking the labels of its nearest data points and going with the majority.Karunya et al. (2025) [3] used KNN alongside Random Forest for classification.

2.1.4 Clustering Approaches (Fuzzy C-Means)

Clustering methods group individuals into similar subgroups without predefined labels. Fuzzy clustering allows degrees of membership to multiple clusters.Wang et al. (2023) [7] used fuzzy clustering on DASS-21 scores to recommend music

2.1.5 Natural Language Processing (NLP)

Some studies, such as Zamani Alavijeh et al. (2025) [5], use NLP and topic modeling to analyze users' tweets. Natural Language Processing, or NLP, is like teaching computers to understand and work with human language in a way that feels natural. It's a field that combines tech and linguistics to help machines make sense of words, sentences, and even entire conversations—whether it's text or speech.

2.2 Related Works

As demonstrated by Rebecchini (2021) [1], music can be utilized as a healing tool in addition to being used for amusement. According to this study, music has an impact on a number of health indicators, including heart rate, motor function, and cognitive activity. It discussed how employing music therapy for mental health can lower the cost of mental health care. In this study, music was offered as a substitute for spoken treatment [1].

Numerous studies investigated the impact of various musical genres on mental health. Rahman et al. (2020) [2] investigated the connections between mood and several musical genres. Based on psychological data collected after an individual began listening to a particular song, a neural network in this study was able to predict an individual's mental health with 98.5% accuracy. By predicting the type of music (99.2% accuracy) a person is listening to based on their electrodermal activity, blood volume pulse, skin temperature, and pupil dilation. This study attempted to establish a relationship between their physiological factors, which affect people's mood and the type of music they are listening to [2].

Another study examined the relationship between various musical genres and mental health. Using the Random Forest algorithm and K-Nearest Neighbour, they were able to predict the musical impact on a person with depression, anxiety, insomnia, and OCD with an accuracy of 85% [3]. In this instance, Karunya et al. (2025) [3] concentrated on the preferred genre or favorite genre of each individual. In this study, SMOTE [8] was utilized to balance the dataset.

Andrew et al. (2024) [4] investigated the connection between people's musical preferences and their mental health, classifying them as happy or displaying poor well-being. They used the Spotify API to gather various musical attributes of songs in order to predict an individual's mental health and well-being based on their playlists. The focus of this study was on personalized song recommendations. They also created a chatbot as an interface to suggest songs according to an individual's WHO well-being index score.This study achieved the highest accuracy of 0.74, a precision of 0.73, and a recall of 0.98 using Random Forest algorithm [4].

In a thorough analysis of the connection between music and mental health, Zamani Alavijeh et al. (2025) [5] presented Twitter-Music PD, a big dataset that uses 8.9 million tweets to examine the relationship between music preferences and mental health. In order to determine a person's preferred music and whether or not they have a diagnosed mental health condition, this study examines their tweets. This author examined the differences in music preferences between those with self-reported mental health disorders and people without such conditions.

Gujar and Reha (2024) [6] applied classical machine learning approaches to understand the relationship between music and the listener's mood. They extracted features from songs labeled with emotional states such as happy, sad, calm, and energetic. In this study, the authors investigated the relationship between specific musical elements and corresponding mood states.

Wang et al. (2023) [7] combined mental health scores (DASS-21) with fuzzy clustering techniques for the purpose of grouping individuals with similar types of mental health conditions. According to this study, no one can be categorized into a single psychological state [7]. Participants in this study were categorized based on the level of their various mental health disorders, including stress, anxiety, and depression. Each group received music recommendations based on the author's domain expertise.

While these studies covered substantial part of the relationship between music and mental health, some critical limitations exist across existing studies. These limitations, identified through a review of prominent works, provided the foundation for the contributions of this thesis.
Most of these studies covered either the emotional content of music or generalized responses without giving attention to individual psychological conditions or mental health conditions. For example, Gujar and Reha [6] applied classical machine learning approaches to understand the relationship between music and the listener's mood using various musical attributes without considering individual psychological condition. A common pattern across some studies is the use musical attributes or low-level audio descriptors (e.g., pitch, rhythm, tempo) to build a relationship between psychological well being (happy, calm or sad) and musical attributes [4] [6]. While these studies are significant to understand relationship between psychological well being and musical attributes, they do not provide insights into user's listening behavior or mental health disorders.
Rahman et al. [2] used physiological signals (e.g., EEG, GSR) to identify emotional responses while listening to a certain type of music. This study is effective in studies with small populations. But for large populations it is not feasible.
Although Rahman et al. [2] Karunya et al. [3] highlight the relationship between mental health states and musical genres, they do not consider the effect of listening frequency of different genres. So the cumulative impact of genres on mental health remains undercover.
Some studies focuses on recommending music based on the similarity between individual's mental health [5] [7], but few attempt to predict musical effect on mental health.
A comprehensive study done by Zamani Alavijeh et al. [5] explains the relationship between musical attributes like it's lyrics and user's mental health disorders, but it

didn't consider the impact of listening to music i'e., whether a person's condition is improving as musical effect or not.

In our study we aim to build more personalized music recommendations for various mental health patients based on their music listening behavior. This research addresses the gap of the existing literature in the ways mentioned below:

- Incorporates different mental health scores (Depression, Anxiety, Insomnia & OCD) as part of the input.
- Frequency of listening to different genres are taken as part of the input while modeling.
- Predicting the musical impact of frequency of listening to different genres on mental health disorders.
- Structuring the dataset for more personalization which makes it useful in mental health support.

Chapter 3

Methodology

This chapter presents the complete methodological pipeline adopted in this research, starting from the acquisition of a relevant dataset to the development of predictive models capable of estimating music's psychological impact. At the heart of this study is the goal of identifying how individuals' genre basaed music listening habits specifically, how often people listen to various music genres alongside their self-reported mental health scores, can be used to predict whether music has a positive, neutral, or negative effect on their well-being and these overcome the limitations of previous studies by introducing more personalized genre-based musical impact predictions for mental health patients.

3.1 Overview

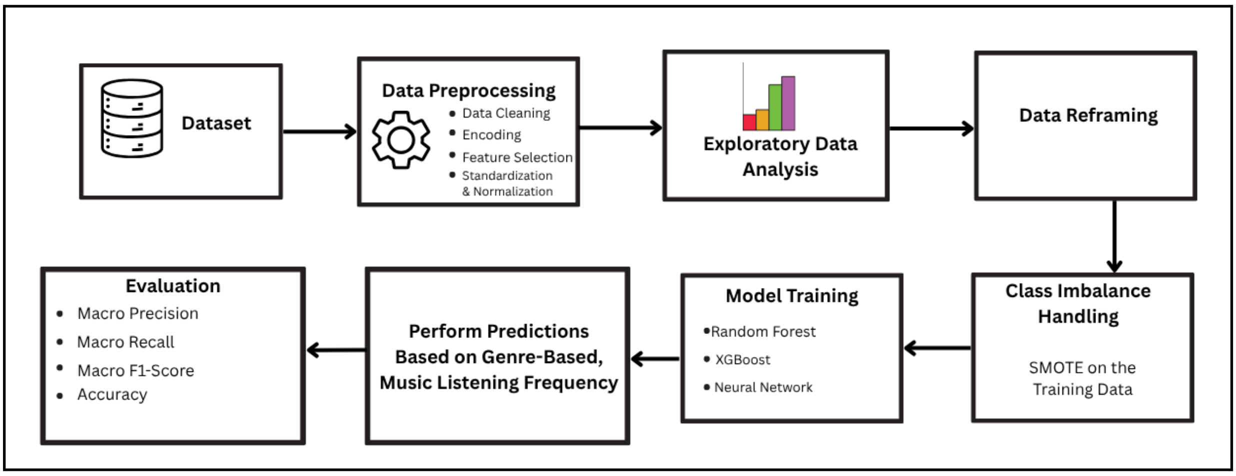

Figure 3.1: Work-Flow Diagram

The process 3.1 begins with the careful selection and preprocessing of a publicly available dataset [9] that captures both behavioral (music preferences) and psychological (mental health indicators) aspects. The data is then reshaped to a long-format structure to isolate the influence of each genre individually, transforming the original participant-level information into genre-level entries. By doing so, the study is able to model more granular relationships between mental health and musical exposure.

Following data preparation, a set of machine learning models—including Random Forest, XGBoost, and a Neural Network—are trained using these transformed inputs. To address class imbalance in the psychological outcomes reported by participants, SMOTE is applied. The models are then rigorously evaluated using k-fold cross-validation and standard classification metrics such as accuracy and macro avg. of precision, recall, and F1-score.

To bridge the gap between research and application, an interactive prediction module is also designed, allowing new users to input their own mental health scores and genre-listening frequencies. The system provides genre-specific predictions that can be potentially used by clinicians or individuals to make more informed decisions about incorporating music into mental health routines.

3.2 Dataset Description

The dataset used in this research was obtained from a publicly available source on Kaggle titled "Music Mental Health Survey" by Catherine Rasgaitis [9]. It comprises responses from 736 participants, who reported their music preferences and mental health conditions through a structured survey.

Frequency	Frequency [Pop]	Frequency [R&B]	Frequency [Rap]	Frequency [Rock]	Frequency	Anxiety	Depression	Insomnia	OCD	Music effects
Never	Very frequently	Sometimes	Very frequently	Never	Sometimes	3	0	1	0	
Never	Sometimes	Sometimes	Rarely	Very frequently	Rarely	7	2	2	1	
Sometime:	Rarely	Never	Rarely	Rarely	Very freque	7	7	10	2	No effect
Never	Sometimes	Sometimes	Never	Never	Never	9	7	3	3	Improve
Never	Sometimes	Very frequently	Very frequently	Never	Rarely	7	2	5	9	Improve
Rarely	Very frequently	Very frequently	Very frequently	Very frequently	Never	8	8	7	7	Improve
Rarely	Rarely	Rarely	Never	Never	Sometimes	4	8	6	0	Improve
Never	Sometimes	Sometimes	Rarely	Never	Rarely	5	3	5	3	Improve
Very frequ	Never	Never	Never	Very frequently	Never	2	0	0	0	Improve
Never	Sometimes	Sometimes	Rarely	Sometimes	Sometimes	2	2	5	1	Improve
Never	Rarely	Rarely	Never	Rarely	Never	7	7	4	7	No effect
Never	Rarely	Rarely	Sometimes	Rarely	Rarely	1	0	0	1	Improve
Never	Sometimes	Sometimes	Rarely	Rarely	Never	9	3	2	7	Improve
Never	Rarely	Never	Very frequently	Never	Never	2	1	2	0	Improve
Sometime:	Sometimes	Never	Never	Sometimes	Rarely	6	4	7	0	Improve
Rarely	Very frequently	Rarely	Sometimes	Sometimes	Rarely	7	5	4	1	Worsen
Never	Never	Sometimes	Very frequently	Never	Rarely	8	8	4	3	Improve
Sometime:	Sometimes	Rarely	Sometimes	Very frequently	Never	5	7	10	0	Improve
Sometime:	Rarely	Sometimes	Very frequently	Very frequently	Never	7	5	0	3	Improve
Rarely	Rarely	Sometimes	Very frequently	Rarely	Never	7	3	0	2	Improve

Figure 3.2: Dataset Demonstration

Relevant fields in the dataset in figure 3.2 include:

- Self-assessed mental health scores: Anxiety, Depression, Insomnia, and OCD (scored on a 0–10 scale).
- Frequency of listening to 16 different music genres (e.g., Rock, Pop, Classical, EDM, Jazz), labeled as Never, Rarely, Sometimes, or Very Frequently.
- Reported psychological effects of music in general—categorized into Improve, No effect, or Worsen.

This dataset was chosen due to its unique combination of behavioral (listening habits) and psychological (mental health condition) data, which provides a valuable foundation for modeling the effect of music on mental well-being.

3.3 Data Preprocessing

To prepare the dataset for modeling, several preprocessing steps were implemented:

- Feature selection: Retained only columns relevant to mental health and music listening behavior.
- Ordinal encoding of genre-listening frequency: 'Never': 0, 'Rarely': 1, 'Sometimes': 2, 'Very Frequently': 3.
- One-hot encoding of categorical genre types.
- Label encoding of the target feature ("Music effects") into numeric values for model training.
- Dropped irrelevant columns such as 'Timestamp', 'Permissions', 'Primary streaming service', and 'BPM' which do not contribute to the analysis.
- Removed records containing null values to ensure data integrity, reducing the number of usable records from 736 to 719.

3.4 Exploratory Data Analysis

To gain a deeper understanding of the dataset and uncover patterns relevant to modeling, several exploratory data analysis techniques were applied. These analyses focused on examining the distribution of music effects, listening habits, and mental health scores, as well as their interrelationships.

3.4.1 Distribution of Music Effects

The "Music effects" variable represents a participant's overall perception of how music affects their mental state. This target variable includes three categories:

- Improve: The participant reports that music improves mood or alleviates symptoms.
- No Effect: Music is perceived as neutral with no noticeable mental impact.
- Worsen: Music is linked to increased stress, anxiety, or other negative effects.

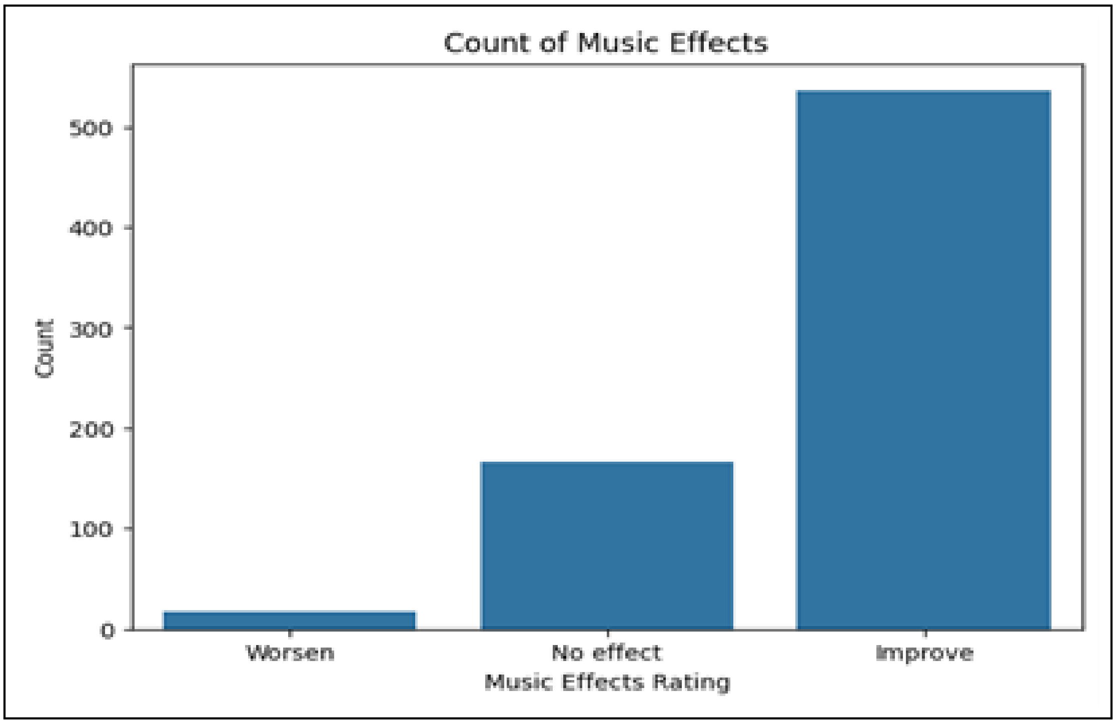

Figure 3.3: Music Effects distribution bar chart

An initial frequency analysis of this variable revealed a pronounced class imbalance:

- Improve: 536 participants
- No Effect: 166 participants
- Worsen: 17 participants

This distribution in figure 3.3 indicates that a majority of respondents experienced positive effects from music, with a significantly smaller group reporting no effect and a very limited number indicating that music worsened their mental health. Such skewed class representation poses a challenge in supervised learning, as models tend to be biased toward the majority class, potentially leading to poor generalization and underperformance in predicting the minority classes.

To mitigate this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was employed during the training phase. SMOTE generates synthetic data points for the underrepresented classes by interpolating between existing minority samples. This approach helps balance the class distribution in the training data, allowing the classification algorithms to learn more representative decision boundaries and improving the reliability of predictions for all three outcome categories.

3.4.2 Mental Health Trends by Favorite Genre

The dataset includes self-reported scores for four key mental health conditions: Anxiety, Depression, Insomnia, and OCD. Each condition is measured on a continuous scale ranging from 0 (no symptoms) to 10 (severe symptoms), allowing for quantitative assessment of participants' psychological states.

To capture a general understanding of how these mental health indicators vary across music preferences, we computed the average score for each mental health variable, grouped by participants' favorite genre. This grouped analysis provides a macro-level view of potential associations between preferred music genres and psychological well-being.

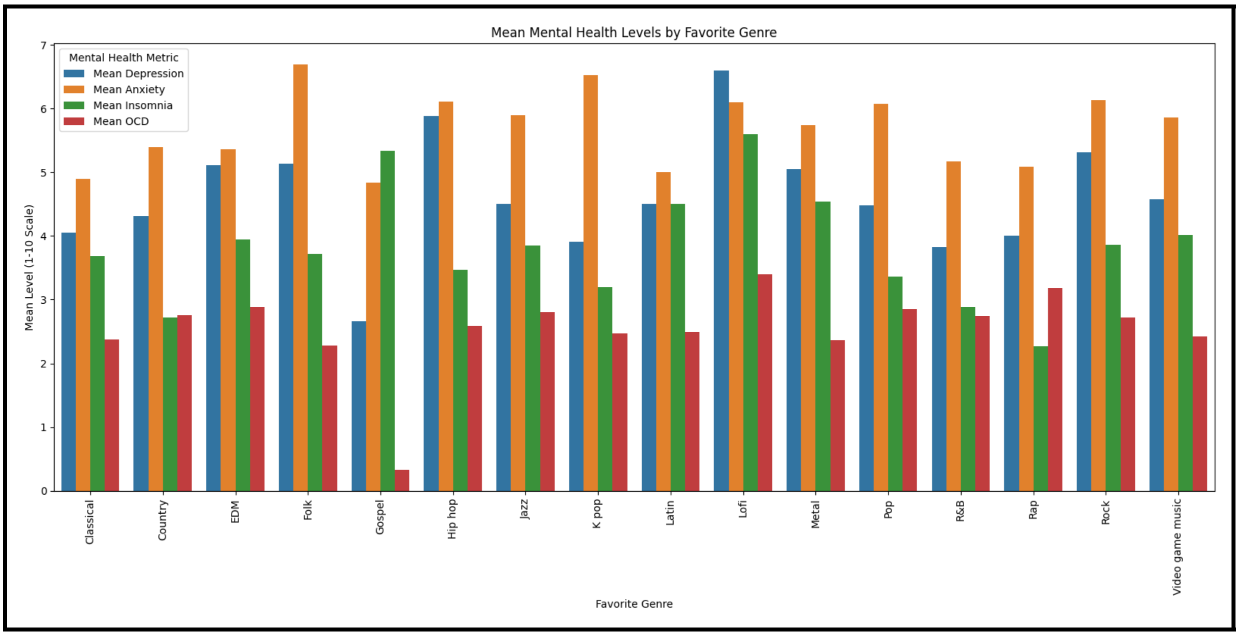

Figure 3.4: Mean mental health levels by favorite genre

The bar chart in figure 3.4 illustrates the mean levels of mental health metrics—depression, anxiety, insomnia, and OCD—across various music genres, providing valuable insights into potential associations between music preferences and mental well-being. Metal and rock fans exhibit the highest levels across all metrics, suggesting a possible link to greater mental health challenges, potentially reflecting the use of these genres as emotional outlets. Conversely, gospel listeners report the lowest levels, indicating a potential protective effect, possibly attributable to its uplifting and spiritual qualities. Classical and country genres are associated with relatively lower levels, hinting at a calming influence, while pop, video game music, EDM, rap, and folk occupy a moderate range, reflecting diverse mental health experiences. These findings suggest opportunities for tailored music therapy interventions and personalized recommendations, such as promoting gospel or classical for emotional support while cautiously integrating metal or rock. However, the observed correlations do not imply causation, and factors such as individual context or pre-existing conditions may influence these patterns. Further research is warranted to explore individual variability, musical elements, and longitudinal effects to fully elucidate the relationship between music preferences and mental health outcomes.

3.4.3 Genre-wise Impact on Perceived Music Effect

To understand how different music genres are perceived in relation to mental well-being, the dataset was analyzed to determine how participants rated the psychological effects of music (Improve, No effect, Worsen) across genres. Since each participant had their genre preferences transformed into long-format rows, we were able to compute genre-level summaries of perceived effects.

Rather than plotting individual boxplots or multiple genre-specific charts, we computed the following aggregated metrics:

- The percentage distribution of “Music effects” (Improve, No effect, Worsen) for each genre.
- A table ranking genres by the proportion of respondents who reported improvement.
- Mean mental health scores grouped by favorite genre (self-reported), to examine whether individuals who prefer certain genres show higher or lower levels of anxiety, depression, insomnia, or OCD.

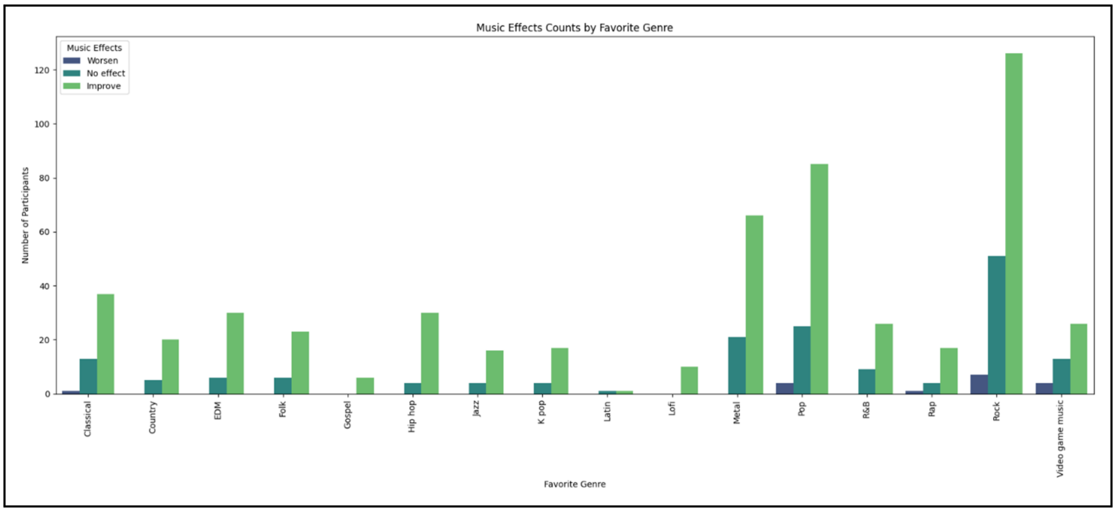

Figure 3.5: Music effect counts by favorite genre

These results in figure 3.5 revealed insightful trends. For instance:

Genres such as Rock, Pop, and Metal were associated with higher “Improve” ratings across participants.

Conversely, genres like Metal (around 20-30 participants with “No effect” or “Worsen”) or EDM (around 20-30 participants with mixed effects) showed more mixed effects, with higher proportions of “No effect” or even “Worsen” ratings, particularly among individuals reporting higher anxiety or depression scores.

3.5 Data Reframing

To boost the analytical power of the dataset and unlock genre-specific predictions, we transformed the original wide-format data into a more dynamic long-format structure. Initially, each respondent's listening frequency across all 16 genres was neatly packed into a single row a tidy setup, but one that obscured the unique impact of each genre on psychological well-being.

The reframing journey involved these key steps:

- Row Expansion: For every participant, we expanded the dataset from one row to 16 rows-one per genre-while carrying over their mental health scores (Anxiety, Depression, Insomnia, and OCD) to each new row. This gave us a richer view of their experience.
- Genre Column Generation: We introduced a new Genre column to clearly tag each row with its associated genre, making the data more navigable.
- Frequency Encoding: We pulled the listening frequency for each genre from the original columns and placed it into a new Frequency column. This was ordinally encoded as follows: 'Never': 0, 'Rarely': 1, 'Sometimes': 2, 'Very Frequently': 3, turning qualitative insights into quantifiable metrics.
- Target Label Alignment: Each expanded row kept the participant's original "Music effects" response (Improve, No effect, or Worsen), linking it directly to the specific genre for a focused analysis.

This transformation injected greater granularity into the dataset, empowering machine learning models to uncover patterns not just from overall listening habits but also from genre-specific interactions. Consequently, our classification system can now explore how the frequency of listening to a particular genre ties to perceived psychological outcomes.

3.6 Feature Engineering

Feature engineering was a pivotal step in this study, transforming raw survey responses into a structured format perfect for machine learning models. Our goal was to unravel how genre-specific music listening habits and mental health profiles shape perceived psychological outcomes, so we carefully crafted features that reflect both the behavioral and psychological sides of the participants' experiences.

3.6.1 Input Features

The final feature set broke down into two key categories:

Mental Health Indicators

We used four continuous variables to capture self-reported mental health status:

- Anxiety (0–10 scale)

- Depression (0–10 scale)
- Insomnia (0–10 scale)
- OCD (0–10 scale)

These were selected for their clinical significance and common use in psychological assessments, offering a lens into how varying mental health challenges might shape music's impact.

Genre Listening Behavior

We included each participant's listening frequency across 16 predefined genres, ordinally encoded as:

- Never = 0
- Rarely = 1
- Sometimes = 2
- Very Frequently = 3

Genre data was handled in two ways: each long-format record tied to a specific genre, and one-hot encoding was applied to keep genre identity as a categorical variable. This let our models pick up on genre-specific effects, moving beyond a generic view of listening habits.

3.6.2 Target Variable

The target feature was the self-reported perceived effect of music, labeled under the "Music effects" variable. This categorical variable was encoded as:

- Improve
- No Effect
- Worsen

These labels reflect participants' personal take on how music influences their mental state, making it the heart of our research aim to predict this outcome based on their listening and psychological profiles.

3.6.3 Significance of Feature Selection

Choosing these features was driven by their theoretical depth and practical value. Mental health scores set a psychological stage, hinting at how individuals' conditions might color their music experiences. Meanwhile, listening frequency tracks a behavioral thread—how often someone dives into a genre—which likely shapes its emotional and cognitive effects.

By focusing on genre-specific frequencies rather than broad listening patterns, our model teases out subtle interactions between mental states and musical inputs. This fine-grained approach boosts the model's interpretability and its real-world potential, especially in clinical settings where tailored, genre-specific recommendations could outshine generic advice.

3.7 Handling Class Imbalance

An important consideration in the model training process was the presence of significant class imbalance in the target variable, "Music effects." Preliminary analysis of the distribution showed that a large majority of participants (536 out of 719) reported that music had an "Improve" effect on their mental health, while 166 participants reported "No effect," and only 17 reported a "Worsen" effect. This imbalance posed a substantial risk of biasing the machine learning models toward the majority class, potentially compromising the reliability and fairness of predictions for minority classes.

To mitigate this issue, SMOTE was employed during the training phase on training data. SMOTE is a widely used resampling method that generates synthetic data points for underrepresented classes by interpolating between existing minority class samples. This approach helps to create a more balanced class distribution without simply duplicating existing records, thereby reducing overfitting and improving model generalization.

The choice of SMOTE was particularly significant for this study, as the goal was to provide meaningful predictions not only for those who experience positive outcomes from music but also for those who are unaffected or potentially harmed. Without proper class balancing, models would likely fail to capture the subtle signals associated with these less common but clinically important outcomes. Ensuring that the model had sufficient exposure to all three classes (Improve, No effect, and Worsen) was essential to support the broader aim of developing personalized, responsible, and evidence-based music therapy recommendations.

3.8 Model Development and Training

To investigate the relationship between music listening behaviors and their psychological effects, we developed and evaluated three supervised machine learning models: Random Forest, XGBoost, and a Neural Network. These models were chosen to reflect a spectrum of algorithmic paradigms, enabling a comparative analysis across ensemble learning, gradient boosting, and deep learning approaches. Each model was trained to classify the perceived psychological impact of music categorized as Improve, No Effect, or Worsen based on individual mental health scores and genre-specific listening frequency.

3.8.1 Model Selection Rationale

The rationale for selecting these models stems from their demonstrated performance in behavioral modeling and mental health-related applications. Their diverse methodological structures provided complementary strengths in handling non-linear data, class imbalance, and the intricate relationships between features.

- Random Forest: This ensemble method constructs multiple decision trees and aggregates their predictions through majority voting. It is particularly valued for its robustness against overfitting, interpretability through feature importance analysis, and efficacy in managing high-dimensional and noisy datasets.
- XGBoost (Extreme Gradient Boosting): Known for its precision and efficiency, XGBoost builds decision trees iteratively, correcting previous errors by optimizing a defined loss function. Its scalability and accuracy make it a leading choice for structured data problems, and it has become a benchmark in machine learning competitions.
- Neural Network: To capture more abstract and non-linear relationships in the data, we implemented a five-layer feedforward neural network. The architecture (512 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 neurons) employed ReLU activation functions, dropout layers for regularization, and a softmax output layer for multiclass classification. The model was trained using the Adam optimizer and categorical crossentropy loss function, with early stopping based on validation performance to avoid overfitting.

All three models received the same input features—mental health indicators, ordinal-encoded genre listening frequencies, and one-hot encoded genre types—ensuring a consistent and unbiased basis for comparison.

3.8.2 Training Pipeline

Prior to training, the dataset was split into training and test sets with an 80/20 ratio using stratified sampling to preserve the original distribution of target classes. Given the evident class imbalance, particularly the underrepresentation of the "Worsen" category, the SMOTE algorithm was applied exclusively to the training data. This step was crucial in generating synthetic samples for the minority classes, thus ensuring balanced representation and enabling the models to learn to recognize all three target outcomes more effectively.

Each model was then trained on the SMOTE-balanced training set:

- The Random Forest model was configured with 800 estimators and a fixed random seed (42) for reproducibility. Its ensemble nature enabled it to detect complex decision boundaries while offering insights into feature relevance.
- The XGBoost model employed a multi-class classification objective (softmax) with log-loss (mlogloss) as the evaluation metric. Its ability to integrate regularization and fine-tune decision boundaries led to consistently superior accuracy and balanced F1 scores.
- The Neural Network model, implemented via TensorFlow's Sequential API, incorporated dropout and early stopping to mitigate overfitting. Though inherently more complex and data-hungry, it showed strength particularly in recognizing the rare "Worsen" class.

3.9 Cross-Validation and Evaluation

To ensure the robustness and generalizability of the predictive models developed in this study, a rigorous evaluation methodology was implemented. Specifically, 5-fold stratified cross-validation was employed. This approach divides the dataset into five equally sized subsets (folds), ensuring that the proportion of target classes (i.e., Improve, No effect, Worsen) is preserved in each fold. In every iteration, one fold is used as the validation set, while the remaining four are used for training. This process is repeated five times, with each fold serving once as the validation set.

Given the observed class imbalance in the target variable (i.e., significantly fewer samples in the 'Worsen' class compared to 'Improve'), each training subset was augmented using SMOTE. This ensured that the models were trained on balanced data during each fold of the cross-validation, thereby mitigating the risk of bias toward the majority class.

The evaluation of model performance was carried out using multiple metrics appropriate for multiclass classification:

- Accuracy: the overall percentage of correctly predicted instances.
- Precision: the ability of the model to return only relevant results for each class.
- Recall: the ability of the model to identify all relevant instances of each class.
- F1-Score (Macro-Averaged): the harmonic mean of precision and recall calculated independently for each class and then averaged. This metric was emphasized due to the imbalanced class distribution, as it provides a balanced view across all outcome categories regardless of sample size.

The evaluation was performed on three classifiers: Random Forest (with 800 estimators), XGBoost, and a Neural Network model with five layers and dropout regularization. During each fold, predictions were generated for the held-out validation set, and the resulting metrics were averaged across the five iterations to provide an estimate of model generalization performance.

Chapter 4

Results and Discussion

This chapter represents the performance results of the three models (Random Forest, XGBoost & Neural Network) which were trained during the method, titled *Leveraging Music Preferences to Assist Psychologists in Tailoring Mental Health Treatments: A Machine Learning Approach*. It discusses the effectiveness of each model in predicting the impact of listening frequency of different genres on mental health conditions. Insights derived from these models are analyzed in the context of mental health and music therapy.

4.1 Quantitative Evaluation

The dataset [9] we used included user-generated genre ratings, psychological assessments across four mental health conditions, and music listening behavior of 736 participants. All three models (Random Forest, XGBoost, and Neural Network) were evaluated using stratified 5-fold cross-validation to maintain balanced class distributions across folds and ensure robust performance assessment. Evaluation metrics used in our study are:

- Macro avg. of precision
- Macro avg. of accuracy
- Macro avg. of recall
- accuracy

Results

Results given by the selected models (RF, XGBoost, Neural Network) are given below:

Table 4.1: RF Evaluation Metrics

	precision	**recall**	**f1-score**
Improve	0.90	0.65	0.76
No effect	0.83	0.61	0.70
Worsen	0.88	0.96	0.92
accuracy			0.87
macro avg	0.87	0.74	0.79

Table 4.2: XGBoost Evaluation metrics

	precision	**recall**	**f1-score**
Improve	0.96	0.82	0.88
No effect	0.90	0.86	0.88
Worsen	0.95	0.97	0.96
accuracy			0.94
macro avg	0.94	0.88	0.91

Table 4.3: Neural Network Evaluation Metrics

	precision	**recall**	**f1-score**
Improve	0.85	0.88	0.86
No effect	0.70	0.80	0.74
Worsen	0.93	0.89	0.91
accuracy			0.87
macro avg	0.83	0.86	0.84

Table 4.4: Proposed Genre Based Model's Performance Comparison

Model	**Accuracy**	**Macro Avg Preision**	**Macro Avg Recall**	**Macro Avg F1-Score**
RF	0.87	0.87	0.74	0.79
XGBoost	0.94	0.94	0.88	0.91
Neural Network	0.87	0.83	0.86	0.84

Table 4.1, 4.2 & 4.3 demonstrate that RF achieved a better F1-score for Worsen class (0.92), but low F1-score for the class Improve (0.76) and No-effect (0.70) and their macro

average is 0.79. Neural Network achieved a better F1-score for Improve (0.86) and Worsen (0.91) classes but a low F1-score for No effect (0.74) class, and their macro average is 0.83. XGBoost achieved the most balanced F1-scores for each class among the three models. From Table 4.4 we can see that, it achieved F1-scores of 0.88, 0.88, and 0.96 for the classes Improve, No Effect, and Worsen, respectively. Additionally, XGBoost's model accuracy is higher than that of RF and Neural Network.

Table 4.5: Existing methods evaluation metrics

No.	Model	Accuracy	Precision	Recall	F1-Score
1.	K-Nearest Neighbor (K-NN)	85.38%	85.70%	85.38%	85.14%
2.	Random Forest	85.00%	85.12%	85.00%	84.74%
3.	Support Vector Machine (SVM)	82.88%	83.04%	82.88%	82.77%
4.	Decision Tree	76.69%	77.14%	76.69%	76.62%
5.	Logistic Regression	66.44%	66.45%	66.44%	65.23%

4.2 Discussion

The result of the study reveals that XGBoost predicts music's effect on a person with a mental health disorder based on their mental health ratings and frequency of listening to different genres, with a highest accuracy of 94% and macro average F1-score of 0.91. It also achieved the highest F1-score of 0.88, 0.88, and 0.96 for individual classes, i.e., Improve, No-Effect, and Worsen, respectively, highlighting a notable correlation between the listening frequency of genres and mental health disorders. The macro F1-scores indicate that minority classes were treated fairly in the test set. Cross-validation confirmed that SMOTE [8] equipped classifiers can produce robust macro F1-scores despite class imbalance. These findings align closely with the research objective of assessing mental states through genre-based music listening behavior. The model shows strong potential in personalized music therapy, but requires refinement for diagnostic accuracy.

From table 4.5 we can see the result from previously done study on this dataset by Karunya et al. [3] which didn't focus on genre based listening frequency as deeply as us and achieved highest accuracy of 85.38% and 85.14% F1-score with K-NN algorithm and 85% accuracy and 84.74% F1-score with RF algorithm. Our model shows strong accuracy using Random Forest (87%) and shows the highest accuracy using XGBoost (94%), highlighting the potential of leveraging listening frequency across different music genres to inform mental health treatments. between listening frequency of different genres and mental health issues.

4.2.1 Comparison with Existing Literature

Comparison between existing literature and our research is demonstrated below in table 4.6:

Table 4.6: Comparison of Existing Approaches with Our Model

Approach	**Main Drawbacks**	**Why Our Model Stands Out**
Body-Signal Methods (e.g., Rahman et al. [2])	Require specialized sensors (GSR, BVP, pupil tracking) and controlled lab settings, limiting real-world scale. Small sample sizes undermine generalizability.	Uses only self-reported mental health scores and music streaming data, which are easy to get and widely available. Can be used anywhere without extra gear, perfect for large-scale, real-world use.
Music-Feature Analysis (e.g., Gujar & Reha [6], Lian & Chen [4])	Focus on basic music traits like speed or pitch but overlook the listener's mental state. Often just group songs by their features instead of predicting how they affect people.	Combines listener's mental health details (like depression or anxiety) with their genre-based music listening habits. Predicts how music impacts each person—whether it helps, has no effect, or makes things worse—instead of just sorting songs.
Favorite-Genre Labeling (e.g., Karunya et al. [3])	Assigns one "favorite genre" per person, ignoring how often they listen.	Breaks down each listener's habits into 16 genres with how often they listen, capturing their preferences better.
Grouping & Rule-Based Suggestions (e.g., Wang et al. [7])	Groups people by their symptoms and uses expert rules to pick genres, not tailoring to individuals with data. Doesn't predict if music will help or harm.	Calculates the chances that a specific genre and listening frequency will help, harm, or do nothing for someone's mental health, making it more personalized.
Social-Media Text Analysis (e.g., Zamani Alavijeh et al. [5])	Guesses mental health and music tastes from social media posts, which can be noisy and unreliable. Hard to turn tweet patterns into practical health advice.	Relies on carefully collected survey data with clear mental health scores and listening habits. Delivers straightforward, useful predictions for clinical or personal mental health support based on genre specific music listening frequency.

Chapter 5

Conclusion

This research investigated the relationship between genre-based music listening behavior and mental health by developing a machine learning system that can predict the psychological impact of music listening behavior. Through structured data collection, data pre-processing, data re-framing, and classification techniques, the study reveals that the listening frequency of different types of music along with combined medical health rating, can be used to forecast whether listening frequency of a particular can improve, worsen or have no effect on an individual's mental health.
Addressing limitations of previous studies, which rely on biometric sensors or couldn't provide personalized genre-based music listening behaviors' impact on people with various mental health issues, our study provides a foundation for real-world application in personalized music therapy and psychological screening. This study explores 16 different genres which covers a comprehensive part of a person's with or without any mental health issues music listening behavior. Psychologists can leverage this study by collecting patients' listening frequency of different music genres, and giving them a mental health issues rating themselves, and evaluating patients' records to ask them to change their music listening frequency of certain genres which has a negative effect on them and keep others genre-based listening behavior as it is. Psychologists' rating of patients' mental health issues will improve this model's accuracy and effectiveness in real life.

While the proposed model is promising in using genre-based music preferences data to assist psychologist in tailoring mental health treatments and achieves good evaluation score in predicting the musical effect of each genre-based listening behavior, there are limitations involved namely, the minimal dataset of this experiment, biasing towards most western genres, lack of demographic data, lack of specialist opinion on the mental health ratings of individuals, class imbalance between improve and worsen class of musical effect.Future research can focus on this limitation to build a more robust tool which can have significant impact on mental health treatments.

5.1 Future works

Several future works can be identified build upon the result of thesis. Future works that can be done are mentioned:

- **Integrating Explainable AI:** Integrating Explainable AI Tools like SHAP or LIME can help unpack how genre-wise listening patterns influence predictions, making the system more transparent and trustworthy for clinicians and therapists.
- **Expert Annotated Mental Health Ratings:** Creating a more robust dataset from psychologists with psychologists ratings of mental health ratings of patients, , allowing the model to reflect clinically validated mental states rather than self-reported assessments alone.
- **Expanding Demographic Diversity:** Training the model on dataset with more diverse data from diverse cultural, geographic, and age groups can improve generalizability.

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